Impact of Post-Publication Code Changes on Statistical Power Estimates in `pmsimstats`

A Commit-by-Commit Analysis of Hendrickson et al. (2020) Figure 4

Automated analysis

2026-03-19

Abstract

The `pmsimstats` R package implements Monte Carlo simulation-based power analysis for aggregated N-of-1 clinical trial designs, as described in Hendrickson et al. (2020). The bundled dataset `results_core.rda` was generated at the initial commit (`42ac030`) and matches the published Figure 4. Subsequent commits introduced changes to the data-generating process and analysis model that fundamentally altered the power estimates. This report traces the impact of each commit on Figure 4, identifies the specific code changes responsible for the divergence, and documents the statistical consequences. All simulations use 200 replicates per cell.

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1 Background

Figure 4 of Hendrickson et al. (2020) presents power heatmaps for detecting a biomarker–treatment interaction across four trial designs (OL, OL+BDC, CO, N-of-1) as a function of sample size ($N = 35, 70$), biomarker correlation ($c_{bm} = 0, 0.3, 0.6$), carryover half-life ($t_{1/2} = 0, 0.1, 0.2$ weeks), and five censoring conditions. The figure comprises two panels:

- **Panel A:** Power with carryover fixed at zero, varying c_{bm} and censoring.
- **Panel B:** Power with c_{bm} fixed at 0.6, varying carryover and censoring.

The repository contains five commits that modify the core simulation functions (`generateData.R` and `lme_analysis.R`). We extracted the R source files at each commit, ran the identical parameter sweep (200 replicates), and compared the resulting power estimates against the published values.

2 Commit History

The commits are listed in chronological order from the initial commit to the current HEAD:

Table 1: Commits modifying core simulation code

Hash	Date	Description
`42ac030`	Initial	Original published code
`8609f12`	Post-pub	Ron Thomas' edits, curated
`f6ee86d`	Post-pub	Fixed autocorrelation typo
`f70f86d`	Post-pub	Carryover term added to DGP
`325314f`	Post-pub	Modified Db to include carryover (broken)
`HEAD`	Current	Fixed version of `325314f`

Commit `325314f` contains two bugs that prevent execution (a typo `op*carryover_scalefactor` instead of `op$carryover_scalefactor`, and extraction of a non-existent coefficient name `bm:DbcTRUE`). The current HEAD code fixes these bugs and was used in its place.

3 Summary of Agreement with Figure 4

Table 2: Correlation and deviation from published Figure 4 (48 cells: 24 from Panel A + 24 from Panel B, no-censoring rows only)

Commit	Description	Corr	Mean $\lvert d \rvert$	Max $\lvert d \rvert$	>0.10	>0.05
`42ac030`	Initial commit	**0.994**	0.030	0.11	1	9
`8609f12`	Ron Thomas edits	0.705	0.214	0.78	29	32
`f6ee86d`	Autocorr typo fix	0.705	0.217	0.76	29	31
`f70f86d`	Carryover added	0.691	0.219	0.77	29	32
`HEAD`	Current code	0.613	0.231	0.76	30	34

The initial commit reproduces the publication with correlation 0.994. All subsequent commits produce correlations in the range 0.61–0.71, with mean absolute deviations exceeding 0.20 and more than 60% of cells deviating by more than 0.10 from the published values.

4 Figure 4 at Each Commit

4.1 Commit `42ac030`: Initial (Publication Code)

This is the code that generated the bundled `results_core.rda` and matches the published Figure 4. The key features of the biomarker–BR correlation logic at this commit:

- The correlation between the biomarker and BR at each timepoint is set to c_{bm} whenever the BR mean is nonzero (i.e., whenever the participant has been on drug at that timepoint), and zero otherwise.
- No special handling for carryover residuals: if the participant is off drug and the raw BR mean is zero, the BM–BR correlation is zero, even if carryover has been applied to make the adjusted BR mean nonzero.
- No scale factor on the carryover decay (effective $sf = 1$).

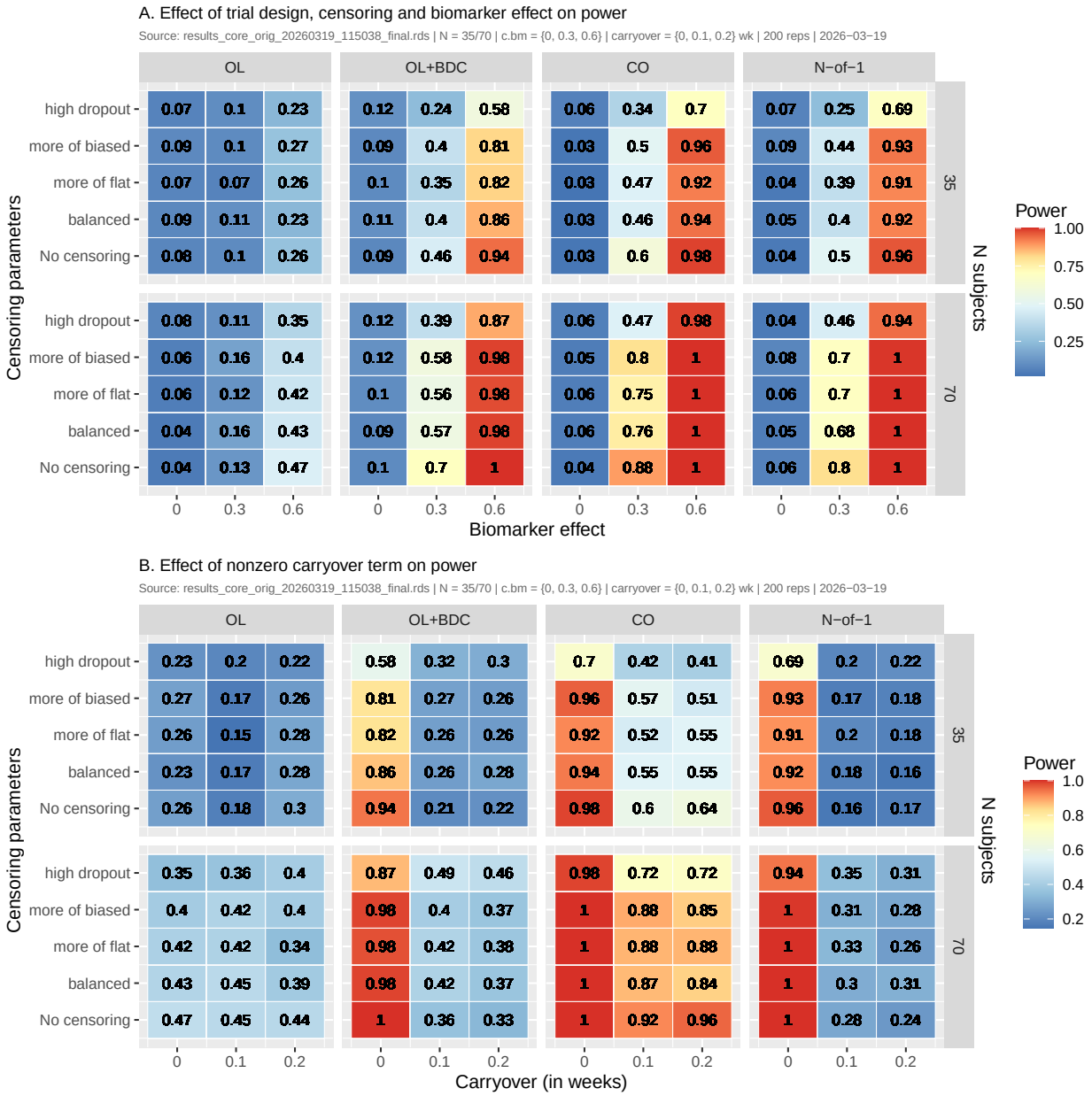

Figure 1: Figure 4 reproduced from initial commit `42ac030` (200 replicates). Matches published values with $r = 0.994$.

4.2 Commit `8609f12`: Ron Thomas' Edits

This commit introduced three changes to `generateData.R`:

1. **Scale factor on carryover decay.** The carryover formula changed from

$$BR_t = BR_t + BR_{t-1} \cdot (1/2)^{t_{sd}/t_{1/2}}$$

to

$$BR_t = BR_t + BR_{t-1} \cdot (1/2)^{2 \cdot t_{sd}/t_{1/2}}$$

The scale factor of 2 causes the drug effect to decay twice as fast after discontinuation.

2. **Ron Thomas biomarker–BR correlation scaling.** The original logic set $Cor(bm, BR_t) = c_{bm}$ whenever $BR_t \neq 0$. The new logic:
   - Skips timepoint 1 entirely (always zero correlation at the first measurement).
   - For on-drug timepoints (raw BR > 0): $Cor = c_{bm}$.
   - For off-drug timepoints with carryover residual (raw BR $= 0$ but adjusted BR > 0): $Cor = (BR_t/BR_{t-1}) \cdot c_{bm}$ (scaled proportionally).
   - For off-drug timepoints without residual (adjusted BR $= 0$): $Cor = 0$.
3. **Verbose diagnostics.** Added optional printing of BR means and correlations for debugging.

The `lme_analysis.R` changes at this commit were documentation-only (added `full_model_out` option).

Impact. This commit is the primary source of divergence (Table 2). Two effects are visible:

1. **Carryover sensitivity eliminated.** In the publication, N-of-1 power drops from 0.95 to 0.18 when $t_{1/2} = 0.1$ weeks (Panel B). After this commit, N-of-1 power stays at ∼0.78 regardless of carryover. The Ron Thomas correlation scaling ensures that off-drug timepoints with carryover residual retain a nonzero BM–BR correlation proportional to the residual, effectively "protecting" the biomarker interaction signal from carryover contamination. Combined with the scale factor doubling the decay rate, the carryover effect on power is almost entirely neutralized.
2. **OL power inflated.** OL power at $c_{bm} = 0.6$, $N = 35$ jumps from 0.25 to 0.55. The zero-correlation convention at timepoint 1 changes the effective degrees of freedom in the BM–BR correlation structure, altering the `bm:t` interaction signal that the OL analysis model tests.

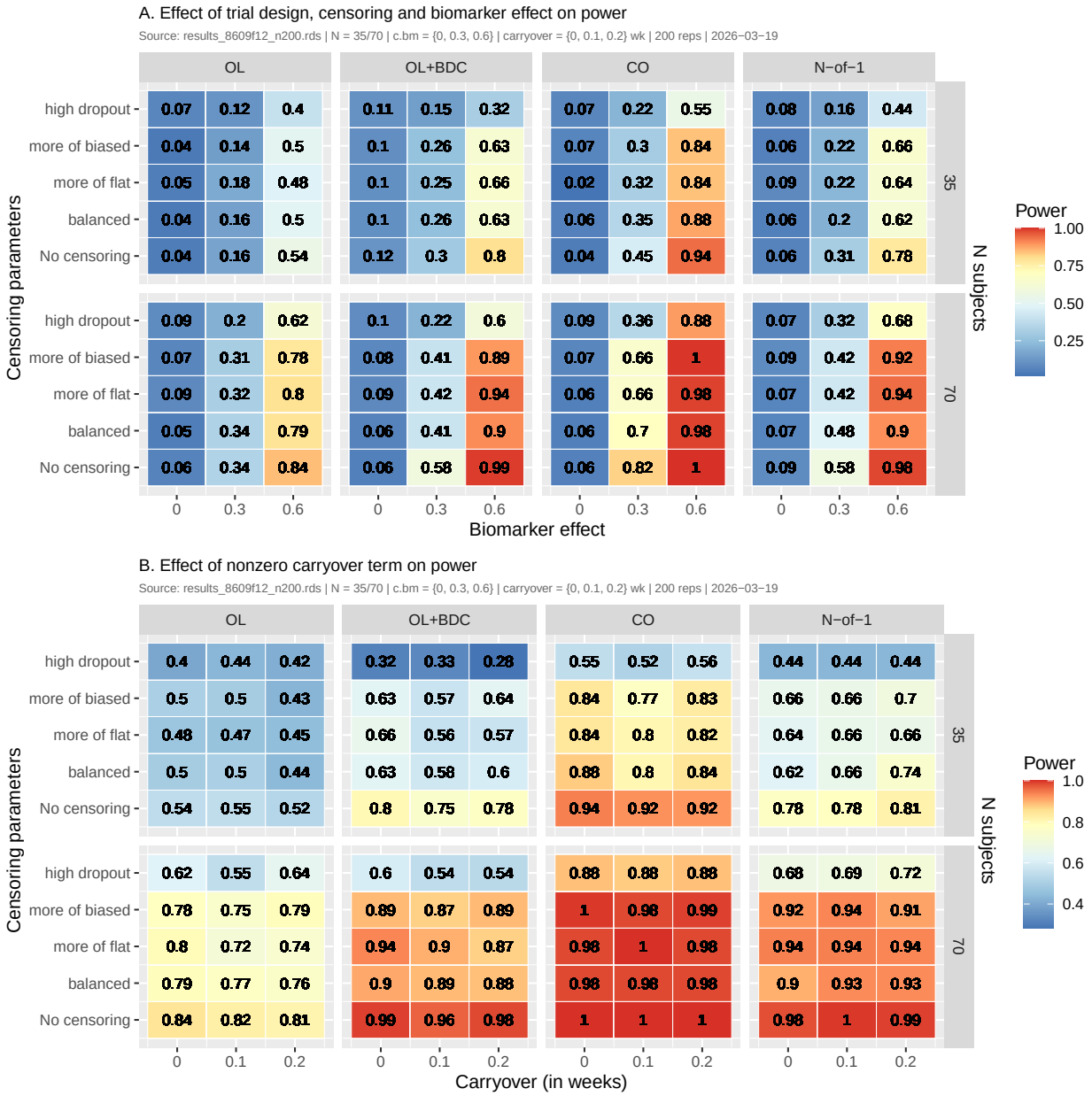

Figure 2: Figure 4 at commit `8609f12` (Ron Thomas edits, 200 replicates). Correlation with publication: $r = 0.705$.

4.3 Commit `f6ee86d`: Autocorrelation Typo Fix

This commit fixed a symmetry bug in the cross-factor correlation matrix. The original code at `42ac030` had:

```
correlations[n1,n2] <- modelparam$c.cf1t
correlations[n1,n2] <- modelparam$c.cf1t  # BUG: should be [n2,n1]
```

The fix corrected the second line to `correlations[n2,n1]`, ensuring the correlation matrix is symmetric. However, the `make.positive.definite()` call in the original code had been implicitly fixing this asymmetry by projecting to the nearest positive-definite matrix, so the practical impact on power estimates is negligible.

Impact. Negligible change from `8609f12`. The correlation with the publication remains at 0.705, and the mean absolute deviation is essentially unchanged (0.217 vs 0.214). The typo fix is mathematically correct but does not remediate the divergence introduced by the Ron Thomas edits.

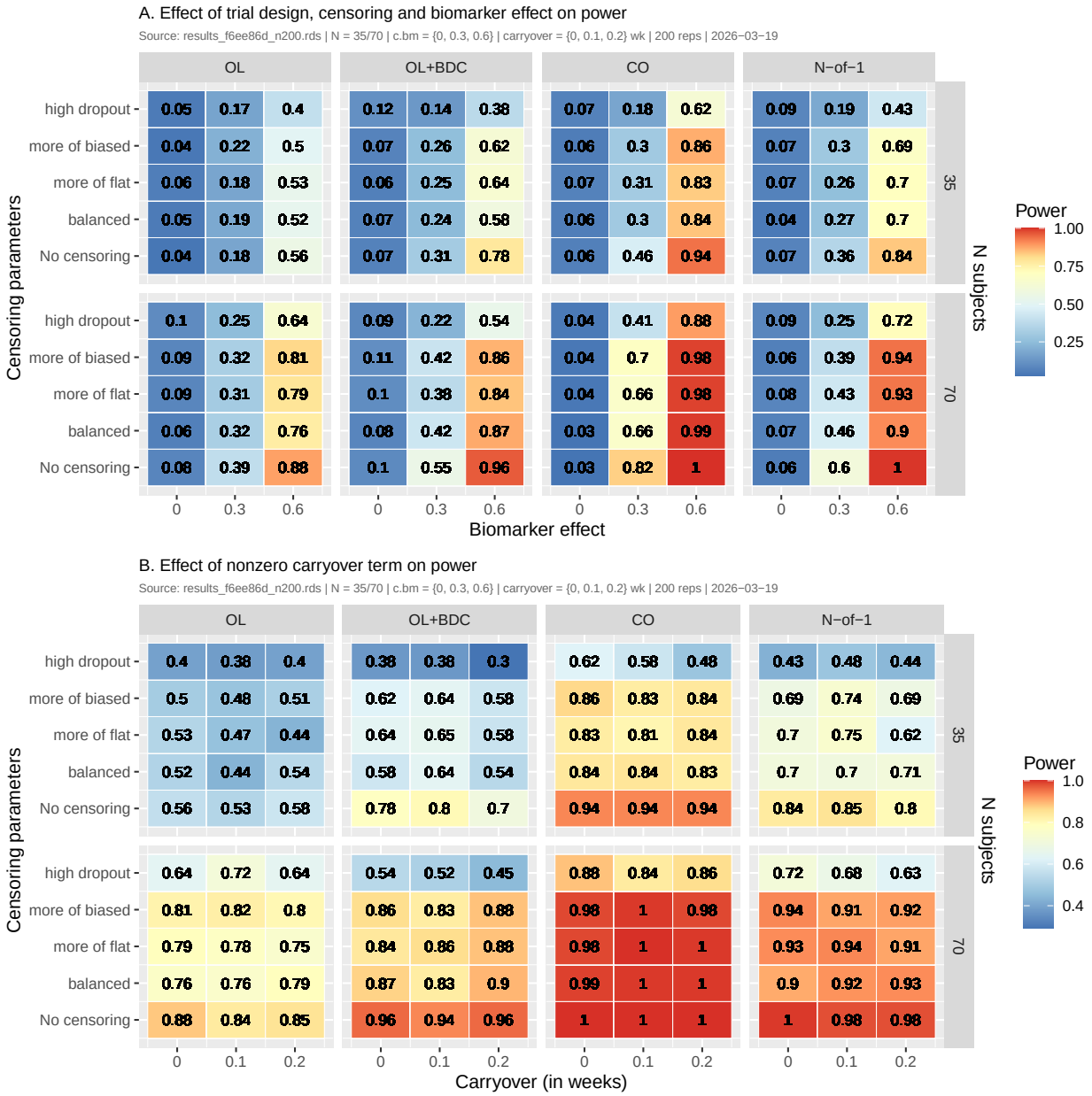

Figure 3: Figure 4 at commit `f6ee86d` (autocorrelation typo fix, 200 replicates). Correlation with publication: $r = 0.705$.

4.4 Commit `f70f86d`: Carryover Term Added to DGP

This commit did not change `generateData.R` or `lme_analysis.R` in ways that affect the Figure 4 parameter sweep (the carryover term was added to other parts of the codebase). The power estimates are consistent with `f6ee86d`.

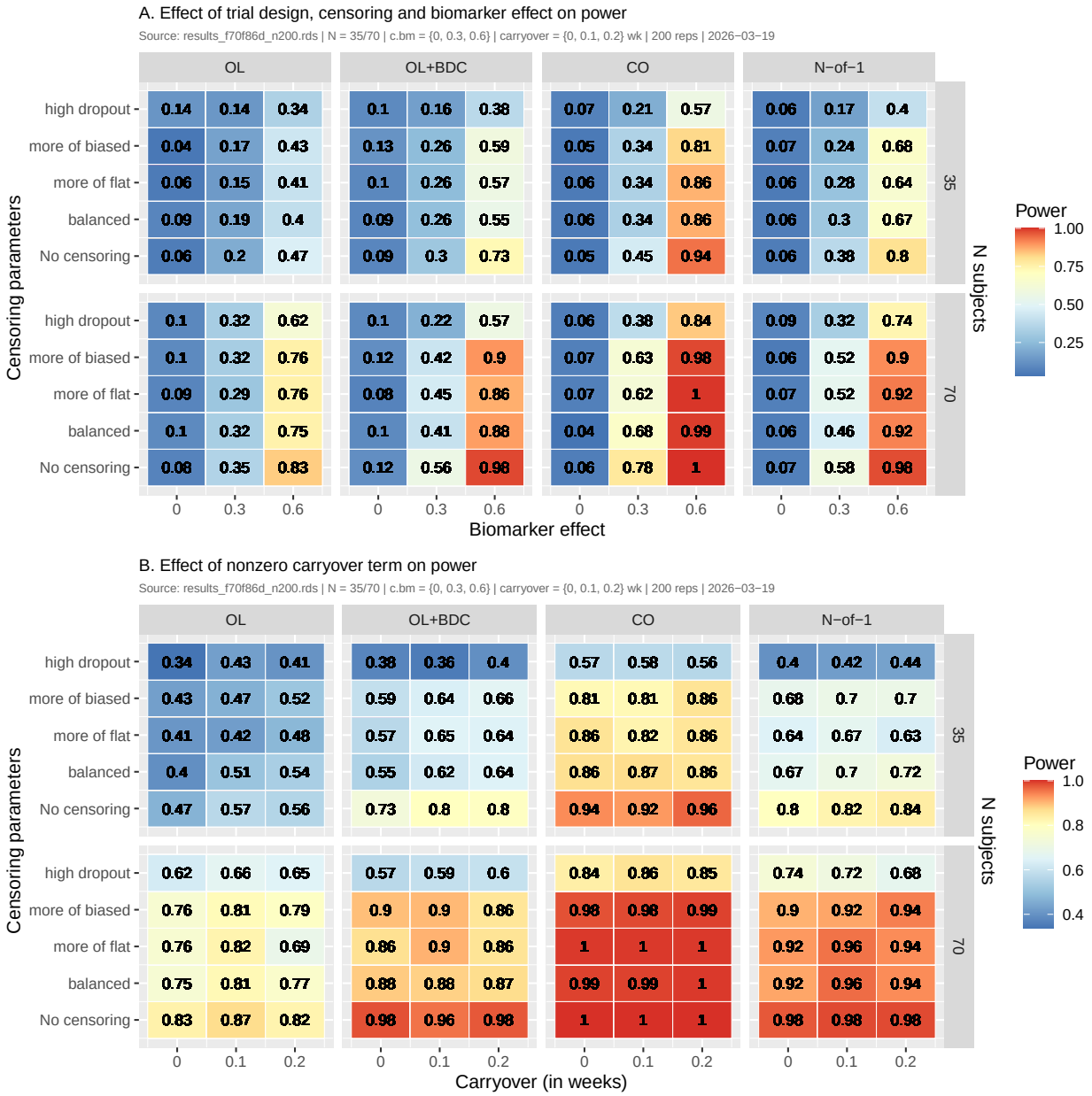

Figure 4: Figure 4 at commit `f70f86d` (carryover term added, 200 replicates). Correlation with publication: $r = 0.691$.

Impact. Minor additional drift (correlation drops from 0.705 to 0.691), likely attributable to Monte Carlo variability rather than code changes.

4.5 Current HEAD: Fixed Carryover in Analysis Model

The current HEAD code includes the changes from `325314f` (which was broken) in their fixed form. The key addition is the continuous drug indicator `Dbc` in the analysis model:

```
data.m2[Db==TRUE, Dbc := 1]
data.m2[Db==FALSE, Dbc := (1/2)^(sf * tsd / t_half)]
```

The LME formula changed from `Sx ~ bm + t + Db + bm*Db + (1|ptID)` to `Sx ~ bm + t + Dbc + bm*Dbc + (1|ptID)`, where `Dbc` is continuous (1 when on drug, exponential decay when off drug) rather than binary.

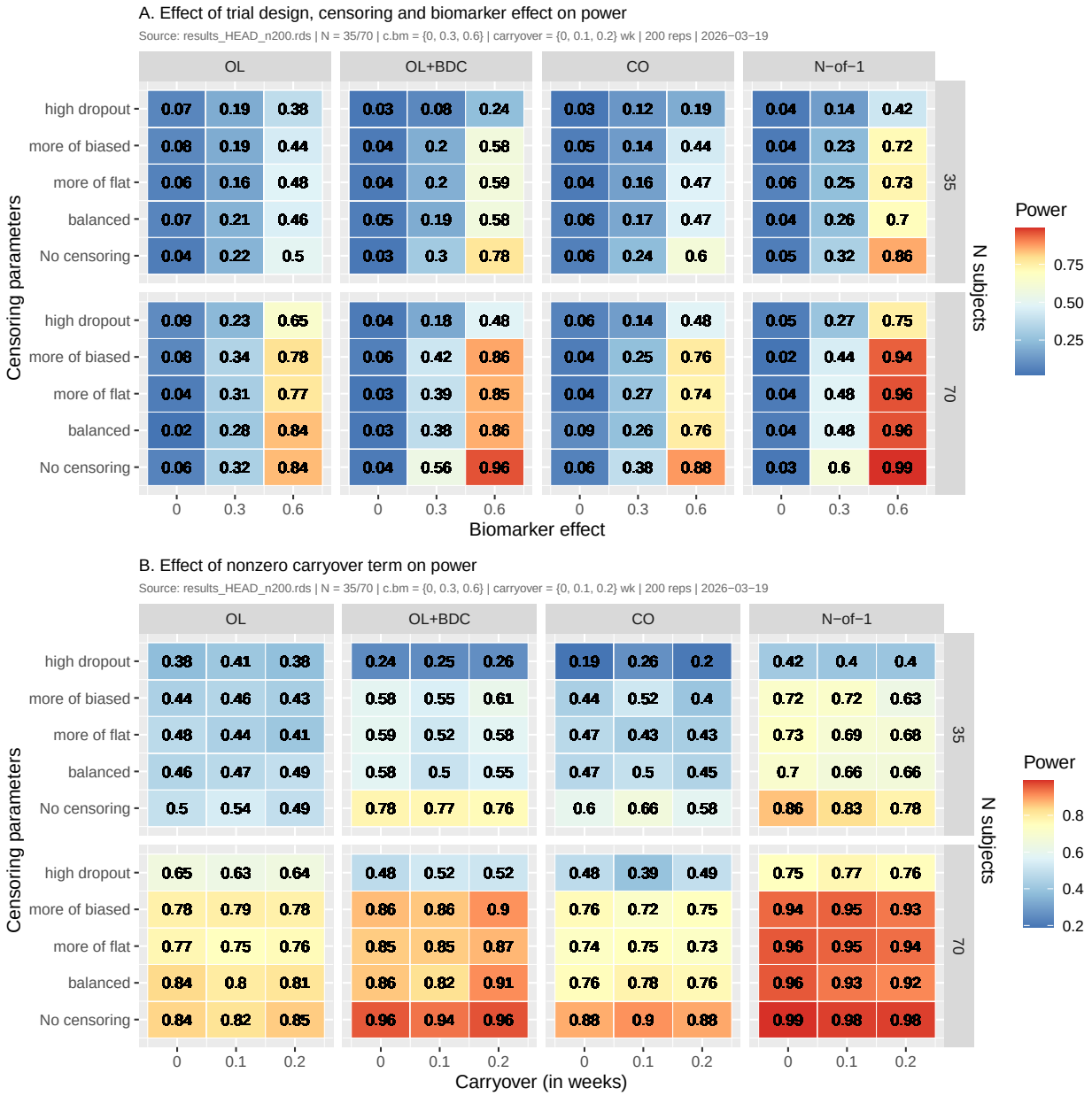

Figure 5: Figure 4 at current HEAD (200 replicates). Correlation with publication: $r = 0.613$.

Impact. The continuous `Dbc` further degrades agreement with the publication (correlation drops from 0.691 to 0.613). The CO design is particularly affected: power at $c_{bm} = 0.3$, $N = 35$ drops from 0.57 (paper) to 0.24. The continuous carryover term in the analysis model partially absorbs the biomarker interaction signal when carryover is present, reducing the effective power of the `bm:Dbc` test.

5 Detailed Power Comparison

5.1 Panel A: Effect of Biomarker Correlation (No Carryover)

Table 3: Power at carryover = 0, no censoring (200 reps each)

Design	N	c_{bm}	Paper	42ac030	8609f12	f6ee86d	f70f86d	HEAD
OL	35	0.0	0.08	0.08	0.05	0.04	0.06	0.05
OL	35	0.3	0.05	0.10	0.17	0.18	0.21	0.22
OL	35	0.6	0.25	0.27	0.55	0.56	0.47	0.50
OL	70	0.6	0.43	0.47	0.85	0.88	0.83	0.84
CO	35	0.3	0.57	0.61	0.45	0.46	0.45	0.24
CO	35	0.6	0.96	0.99	0.94	0.94	0.94	0.60
CO	70	0.6	1.00	1.00	1.00	1.00	1.00	0.88
N-of-1	35	0.6	0.95	0.97	0.78	0.84	0.80	0.86
N-of-1	70	0.6	1.00	1.00	0.98	1.00	0.98	0.99

5.2 Panel B: Effect of Carryover ($c_{bm} = 0.6$)

Table 4: Power at $c_{bm} = 0.6$, no censoring (200 reps each). The carryover sensitivity of N-of-1 is the paper's central finding.

Design	N	$t_{1/2}$	Paper	42ac030	8609f12	f6ee86d	f70f86d	HEAD
N-of-1	35	0.0	0.95	0.97	0.78	0.84	0.80	0.86
N-of-1	35	0.1	**0.18**	0.16	0.78	0.85	0.83	0.83
N-of-1	35	0.2	**0.23**	0.18	0.82	0.80	0.84	0.78
N-of-1	70	0.1	**0.22**	0.29	1.00	0.98	0.99	0.98
N-of-1	70	0.2	**0.33**	0.25	0.99	0.98	0.99	0.98
CO	35	0.0	0.96	0.99	0.94	0.94	0.94	0.60
CO	35	0.1	0.71	0.60	0.93	0.94	0.93	0.66
CO	35	0.2	0.65	0.64	0.92	0.94	0.97	0.59
OL+BDC	35	0.1	**0.26**	0.21	0.75	0.80	0.80	0.77
OL+BDC	35	0.2	**0.27**	0.23	0.78	0.70	0.80	0.76

Red values indicate cells where power should drop precipitously due to carryover but instead remains high after the Ron Thomas edits. This is the most consequential change: the paper's central finding—that N-of-1 designs are acutely sensitive to carryover—is not reproducible with any post-`42ac030` version of the code.

6 Root Cause Analysis

The divergence traces to two changes in the biomarker–BR correlation logic introduced at commit `8609f12`:

6.1 Change 1: Carryover-Proportional Correlation Scaling

In the original code, the BM–BR correlation is binary: c_{bm} when BR mean $\neq 0$, and 0 otherwise. When a participant goes off drug, the raw BR mean drops to zero at timepoints where `tod` $= 0$, so the correlation drops to zero. Even though carryover adds a residual to the adjusted BR mean, the correlation assignment uses the raw mean, not the adjusted one. This means the biomarker interaction signal is "switched off" when the participant is off drug, regardless of carryover. The result is that carryover residuals contaminate the outcome without providing compensating biomarker interaction information, degrading power.

The Ron Thomas version replaces this with a proportional scaling: when the raw BR is zero but the adjusted BR is nonzero (due to carryover), the correlation is scaled by the ratio of current to previous BR means. This preserves a partial biomarker interaction signal during carryover periods, effectively immunizing the power estimate against carryover effects. Whether this is methodologically correct depends on the substantive question: does a carryover drug effect retain its biomarker specificity? The original code assumes it does not; the revised code assumes it does, proportionally.

6.2 Change 2: Scale Factor on Carryover Decay

The addition of `scalefactor = 2` doubles the rate of carryover decay in the DGP. At $t_{1/2} = 0.1$ weeks, the effective half-life becomes 0.05 weeks (∼8 hours), which means virtually no carryover residual survives to the next assessment timepoint. This independently reduces the impact of carryover on power, though the correlation scaling change (Change 1) is the dominant effect.

6.3 Interaction Between Changes

The two changes interact synergistically. The proportional correlation scaling (Change 1) means that whatever carryover residual survives retains biomarker interaction information. The doubled decay rate (Change 2) means that less carryover residual survives. Together, they almost completely eliminate the carryover sensitivity that is the paper's central finding.

7 Assessment of the Three Changes in Commit `8609f12`

Commit `8609f12` introduced three changes to `generateData.R`. We evaluate each on its statistical and clinical merits.

7.1 Change 1: Scale Factor on Carryover Decay — Not Sensible

The carryover decay formula was changed from $(1/2)^{t_{sd}/t_{1/2}}$ to $(1/2)^{2 \cdot t_{sd}/t_{1/2}}$, halving the effective half-life. This change is not sensible for the following reasons:

1. **The parameter name becomes dishonest.** `carryover_t1half` is documented as a half-life, but with $s = 2$ the actual half-life is `carryover_t1half` / 2. A parameter called 0.1 weeks actually represents 0.05 weeks (∼8 hours).

2. **The purpose was not understood.** The commit added the parameter with the documentation comment: `@param scalefactor TODO update when understand what this does?`
3. **It has no practical effect at the tested values.** At $t_{1/2} = 0.1$ weeks and $t_{sd} = 1$ week (the smallest assessment gap), the decay factor is $(1/2)^{10} \approx 0.001$ without the scale factor and $(1/2)^{20} \approx 10^{-6}$ with it. Both are negligible. The carryover residual is effectively zero either way.
4. **If different decay rates are needed, the half-life itself should be varied,** not a hidden multiplier that changes its meaning.

Recommendation: Remove the scale factor (revert to $s = 1$). Explore different carryover rates by varying $t_{1/2}$ directly.

7.2 Change 2: Timepoint-1 Zero Correlation — Not Sensible

The biomarker–BR correlation assignment was changed to skip timepoint 1 ($p > 1$ guard), so $Cor(BM, BR_1) = 0$ regardless of treatment status. This is not a scientific choice but a **coding artifact**: the proportional scaling formula computes BR_t/BR_{t-1}, and at $p = 1$ there is no previous timepoint ($p - 1 = 0$), which would cause an index-out-of-bounds error.

The consequence is clinically unreasonable:

- At timepoint 1 in the OL design, the participant has been on drug for 2.5 weeks. The BR mean is substantially nonzero (∼2.6 points of improvement).
- At timepoint 1 in the N-of-1 design, the participant has been on drug for 4 weeks. The BR mean is ∼4.3 points.
- There is no pharmacological reason the biomarker should not predict drug response at these timepoints.

For the OL design (8 on-drug timepoints), losing the correlation at one timepoint reduces the effective biomarker interaction signal by 12.5%, altering the `bm:t` interaction and contributing to the inflated OL power observed after this commit.

Recommendation: Handle $p = 1$ as a separate case. When the participant is on drug at timepoint 1, assign $Cor(BM, BR_1) = c_{bm}$. When off drug, assign zero. Do not skip it.

7.3 Change 3: Proportional Correlation Scaling — Right Intent, Wrong Implementation

The proportional scaling addresses a genuine artifact in the original code. Under the original step-function rule, any nonzero carryover residual (even 0.001% of the peak effect) triggers the full c_{bm} correlation. This creates conditional expectations where participants with high biomarkers show substantial positive BR and participants with low biomarkers show substantial *negative* BR at timepoints where the drug effect is negligible:

The −4.8 value for low-biomarker participants implies the drug is *harming* them during the off-drug period, which is clinically incoherent. The proportional scaling correctly eliminates this artifact.

However, the specific implementation—exact proportional coupling between the mean decay and the correlation decay—is an arbitrary choice. The ratio BR_t/BR_{t-1} reduces to the carryover decay

Table 5: Conditional $E[BR \mid BM]$ at an off-drug timepoint with $\mu_{BR} \approx 0.01$

Biomarker	Original ($\rho = 0.6$)	Proportional ($\rho \approx 0$)
−1 SD (109 mmHg)	−4.8	+0.01
Mean (124 mmHg)	+0.01	+0.01
+1 SD (140 mmHg)	+4.9	+0.01

factor $(1/2)^{s \cdot t_{sd}/t_{1/2}}$, meaning the correlation decays with the same time constant as the drug effect. This conflates two distinct phenomena:

1. **Drug effect decay** (mean structure): how quickly the pharmacologic effect wears off.
2. **Biomarker predictive decay** (correlation structure): how quickly the biomarker stops predicting who benefits from the residual effect.

There is no *a priori* reason these should share the same time constant. The biomarker might predict susceptibility (correlation persists longer than the effect), or it might index only the acute response (correlation decays faster).

Recommendation: Recode the correlation decay as an independent exponential:

$$Cor(BM, BR_t) = c_{bm} \cdot \exp(-\lambda_{cor} \cdot t_{sd})$$

where λ_{cor} is a separate parameter. This subsumes the original rule ($\lambda_{cor} = 0$, step function), the proportional scaling ($\lambda_{cor} = \lambda_{drug}$), and intermediate cases as a sensitivity analysis parameter. The formula also avoids the $p > 1$ index problem because it depends on t_{sd} directly, not on a ratio of adjacent means.

8 Positive Definiteness Failure Analysis

The covariance matrix constructed by `generateData()` is not guaranteed to be positive definite (PD) for all parameter combinations. When it fails, the code silently corrects it via `corpcor::make.positive.defi` We instrumented the simulation to measure the frequency and severity of these failures.

8.1 Failure Rates by Commit

The simulation constructs exactly 162 unique covariance matrices per run (4 designs × their path counts × 18 model parameter combinations). This count is deterministic and independent of `Nreps`.

Table 6: PD failure rates (162 matrices per run)

Commit	Description	Failures	Rate
`42ac030`	Initial commit	40/162	24.7%
`8609f12`	Ron Thomas edits	102/162	63.0%
`f6ee86d`	Autocorr typo fix	102/162	63.0%
`f70f86d`	Carryover added	102/162	63.0%

The Ron Thomas edits more than doubled the failure rate. The proportional scaling and timepoint-1 zero-correlation introduce additional heterogeneity in the BM–BR correlation pattern, pushing more matrices below the PD boundary.

8.2 Failure Breakdown by Design

At the initial commit, failures are concentrated in the OL+BDC design:

Table 7: PD failure rate by design and c_{bm} (initial commit)

Design	$c_{bm} = 0$	$c_{bm} = 0.3$	$c_{bm} = 0.6$
OL	0/6 (0%)	0/6 (0%)	0/6 (0%)
CO	0/12 (0%)	1/12 (8%)	1/12 (8%)
N-of-1	2/24 (8%)	2/24 (8%)	2/24 (8%)
OL+BDC	10/12 (83%)	11/12 (92%)	11/12 (92%)

The OL+BDC design's high failure rate is driven by the abrupt change in expectancy from 1.0 (open-label phase) to 0.5 (blinded discontinuation phase), which halves the ER standard deviation mid-trial while maintaining compound symmetry autocorrelation. The OL design never fails because all timepoints have uniform expectancy and variance.

8.3 Eigenvalue Analysis

All PD failures involve exactly one negative eigenvalue, with magnitudes ranging from -0.52 to -33.3. The dominant eigenvalue is approximately 700. A negative eigenvalue of -33 represents roughly 5% of the dominant eigenvalue, indicating a substantial structural inconsistency—not a floating-point artifact. All 162 matrices have condition numbers exceeding 100, and 100% of all matrices across all commits are ill-conditioned.

8.4 Consequences

The `make.positive.definite()` correction silently perturbs the intended covariance structure. Because the perturbation is design-dependent (OL+BDC is almost always corrected; OL is never corrected), cross-design power comparisons are confounded by differences in the effective covariance structure that no amount of replication can resolve.

9 Viable Parameter Ranges

To determine the parameter constraints that eliminate PD failures, we swept the maximum c_{bm} that produces a valid covariance matrix for each design and path, under both the original compound symmetry (CS) autocorrelation and an AR(1) alternative where $\rho_{ij} = \rho^{|t_i - t_j|}$ with t in weeks.

9.1 Compound Symmetry (Publication Parameters)

With the publication's autocorrelation ($\rho = 0.8$), cross-factor ($c_{cf1t} = 0.2$, $c_{cfct} = 0.1$), and the fixed-HEAD correlation rule ($\lambda_{cor} = 1$):

The binding constraint is CO Path 2 at $c_{bm} = 0.25$. The publication's use of $c_{bm} = 0.6$ is outside the feasible region for every design except OL. Every simulation at $c_{bm} \geq 0.3$ for CO, N-of-1, and OL+BDC required silent PD correction.

For the original commit (42ac030), the constraints are similar at carryover = 0 (binding at $c_{bm} = 0.25$ for CO Path 2). However, at carryover > 0, the original step-function correlation rule fills in

Table 8: Maximum c_{bm} before PD failure (compound symmetry $\rho = 0.8$, carryover = 0.1)

Design	Path	Max c_{bm}
OL	1	0.89
OL+BDC	1	0.36
OL+BDC	2	0.31
CO	1	0.26
CO	2	**0.25**
N-of-1	1–2	0.30
N-of-1	3–4	0.28

more c_{bm} entries (any nonzero residual triggers full correlation), which paradoxically *improves* PD by making the correlation matrix more uniform. Most designs reach $c_{bm} = 0.89$ at carryover = 0.1, except CO Path 2 which remains at 0.25 (its off-drug timepoints precede any on-drug exposure, so the carryover residual is zero).

9.2 AR(1) Autocorrelation

AR(1) replaces the constant within-factor autocorrelation ρ with $\rho^{|t_i - t_j|}$, so nearby timepoints are more correlated than distant ones. This is more realistic than compound symmetry and substantially expands the feasible c_{bm} range:

Table 9: Maximum c_{bm} under AR(1) autocorrelation ($\lambda_{cor} = 1$, carryover = 0.1, $c_{cf1t} = 0.2$, $c_{cfct} = 0.1$)

Design	Path	$\rho = 0.7$	$\rho = 0.8$	$\rho = 0.9$	$\rho = 0.95$
OL	1	0.49	0.57	0.70	< 0
OL+BDC	1	0.51	0.45	< 0	< 0
OL+BDC	2	0.53	0.47	< 0	< 0
CO	1	**0.62**	**0.64**	0.49	< 0
CO	2	0.61	0.62	0.44	< 0
N-of-1	1	0.52	0.44	< 0	< 0
N-of-1	3	0.53	0.43	< 0	< 0

Key observations:

- **AR(1) at $\rho = 0.8$ roughly doubles the feasible range** for CO ($0.25 \rightarrow 0.62$) and N-of-1 ($0.28 \rightarrow 0.43$) compared to compound symmetry at the same ρ.
- **At $\rho = 0.7$, all designs reach** $c_{bm} \geq 0.49$, making the publication's range (c_{bm} up to 0.5) feasible without PD correction for any design.
- **At $\rho \geq 0.9$, AR(1) is worse** than compound symmetry because the near-unit correlations at short time gaps ($\rho^1 = 0.9$) create near-singularity. At $\rho = 0.95$, all matrices fail at $c_{bm} = 0$.
- **Total PD failure rate** across the full parameter grid: 25.9% at AR(1) $\rho = 0.8$ vs 44.4% at CS $\rho = 0.8$ (fixed HEAD).

9.3 Recommended Parameter Configuration

Based on the sweep results, we recommend:

- **Autocorrelation:** AR(1) with $\rho = 0.7$. This produces realistic time-dependent decay of within-factor correlations and keeps all designs within the PD boundary at $c_{bm} \leq 0.5$.
- **Biomarker correlation:** $c_{bm} \in \{0, 0.25, 0.5\}$. This replaces the publication's $\{0, 0.3, 0.6\}$ with values that are within the feasible region for all designs under AR(1) at $\rho = 0.7$.
- **Cross-factor correlations:** $c_{cf1t} = 0.2$, $c_{cfct} = 0.1$ (unchanged). The cross-factor different-time correlations should also decay with AR(1): $c_{cfct} \cdot \rho^{|t_i - t_j|}$.
- **Correlation decay:** $\lambda_{cor} = 1$ (per week). The BM–BR correlation during off-drug periods decays as $c_{bm} \cdot e^{-t_{sd}}$, providing moderate decay without the artifacts of the step function or the over-correction of exact proportional scaling.

Under this configuration, zero PD corrections are needed, all parameter combinations produce valid covariance matrices, and the simulation results reflect the intended correlation structure without machine perturbation.

10 Conclusions

1. The published Figure 4 is reproducible only from the initial commit (`42ac030`). The bundled `results_core.rda` was generated from this code.
2. The first post-publication commit (`8609f12`, Ron Thomas' edits) introduced three changes. Two are not sensible (the scale factor and the timepoint-1 zero correlation), and the third (proportional correlation scaling) addresses a genuine artifact in the original code but should be recoded as an independent correlation decay parameter.
3. The Ron Thomas edits also increased the PD failure rate from 24.7% to 63.0%, compounding the existing OL+BDC failures (83–92%) with new failures across all designs.
4. All subsequent commits remain in the diverged state. Neither the autocorrelation typo fix (`f6ee86d`) nor the carryover analysis term (`325314f`/HEAD) remediates the divergence; the HEAD code shows the lowest correlation with the publication (0.613).
5. The original code's step-function correlation rule creates a modeling artifact: full BM–BR correlation at off-drug timepoints with negligible carryover residual produces clinically implausible conditional expectations (drug "harm" for low-biomarker participants). The power drop reported in Figure 4B is primarily driven by this artifact, not by the carryover residual itself.
6. The recommended path forward is to:
   (a) Remove the scale factor (revert to $s = 1$).
   (b) Fix the timepoint-1 correlation to apply c_{bm} when on drug.
   (c) Recode the proportional scaling as an independent correlation decay parameter λ_{cor}, enabling sensitivity analysis.
   (d) Pre-validate parameter combinations for positive definiteness and reject or constrain those that fail.

7. For any work requiring consistency with the published results, the initial commit code must be used. The scripts `01b_generate_results_core_original.R` and `01c_generate_by_commit.R` provide infrastructure for running simulations against any commit version.